SIMATS ENGINEERING
CSA6303- Threat Intelligence and Network Security
CO3-AT1-Threat Intelligence Report

Assessment Questions

Subject	Threat Intelligence	**Total Marks**	50
Questions	5 (all compulsory)	**Marks per Question**	10
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Instructions: Answer all questions. Each question carries 5 marks. Support answers with examples and diagrams where appropriate.

CO3: Perform threat analysis using intelligence models and utilize industry tools for monitoring and detection.

Code of Conduct:

I B.Hari Sai/192311098 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of Student :B.Hari Sai

Questions :

1. **Threat Hunting Techniques:** An organization suspects that an attacker has gained unauthorized access to its internal network. The security team observes unusual login times, unexpected processes, and connections to unfamiliar external servers. Describe the proactive threat-hunting techniques you would use to investigate the activity and identify potential threats. [10]

2. **SIEM Tools and Security Monitoring :** A company implements a Security Information and Event Management (SIEM) system to monitor its network. The system generates multiple alerts about suspicious login attempts and unusual outbound connections. Explain how the SIEM system can collect and analyze security events, correlate alerts, and help security analysts prioritize potential threats. [10]
3. **Security Monitoring and Anomaly Detection:** A company's security monitoring dashboard identifies a sudden increase in outbound network traffic during non-working hours. The activity originates from a database server that normally transfers only small amounts of data. Explain how you would investigate the anomaly, identify possible data exfiltration, and recommend suitable monitoring and prevention measures. [10]
4. **Tools Used in the IT Industry for Threat Hunting:** An IT organization suspects that malware has infected several employee workstations. The security team has access to SIEM software, endpoint detection tools, network traffic analyzers, and vulnerability scanners. Select suitable tools for investigating the incident and explain how each tool can help identify, analyze, and contain the threat. [10]
5. **Integrated Threat Hunting and Incident Reporting:** A company detects suspicious user logins, connections to a malicious domain, and an unknown process running on an employee workstation. The security team uses Splunk or the ELK Stack to analyze logs and a threat intelligence platform to verify suspicious indicators. Describe the complete threat-hunting process, including data collection, log correlation, threat investigation, malware analysis if required, risk assessment, and preparation of a threat-hunting report with recommended remediation actions. [10]

ANSWERS:

1. Threat Hunting Techniques [10 Marks]

Introduction

Threat hunting is the proactive process of searching for hidden cyber threats that may have bypassed traditional security tools. In this scenario, unusual login times, unexpected processes, and connections to unfamiliar external servers may indicate that an attacker has gained unauthorized access.

Log Analysis

Examine authentication and system logs to identify logins at unusual times, repeated failed attempts, access from unfamiliar locations, and unexpected privilege changes. Compare these events with normal user activity to identify suspicious patterns.

Endpoint Monitoring

Inspect running processes, startup programs, scheduled tasks, and recently modified files on affected computers. Identify unknown programs or processes consuming unusual resources, and check their file hashes and digital signatures.

Network Traffic Analysis

Review network connections, destination IP addresses, ports, protocols, and traffic volumes. Look for repeated communication with unfamiliar servers, unusual data transfers, or connections to known malicious infrastructure.

Indicators of Compromise (IOCs)

Search for suspicious IP addresses, domain names, file hashes, and registry changes. Compare these indicators with trusted threat-intelligence sources to determine whether they are associated with known attacks.

Behavioral Analysis

Establish a baseline of normal user and system behavior. Investigate deviations such as logins outside working hours, access to sensitive files, unusual administrative commands, or large data transfers.

Threat Investigation and Containment

Build a timeline of the suspicious activity and determine which accounts and devices may be affected. If compromise is confirmed, isolate affected devices, disable or reset compromised accounts, and block malicious connections while preserving evidence.

Conclusion

Threat hunting combines log analysis, endpoint inspection, network monitoring, and threat intelligence to uncover hidden attacks early and reduce the risk of data theft or system damage.

2. SIEM Tools and Security Monitoring [10 Marks]

Introduction

Security Information and Event Management (SIEM) systems collect security events from different parts of an organization, analyze them centrally, and help analysts detect and investigate possible attacks.

Data Collection

A SIEM collects logs from servers, workstations, firewalls, routers, applications, identity systems, and endpoint security tools. Centralizing this information gives analysts a broader view of activity across the organization.

Log Normalization and Analysis

The system converts different log formats into a common structure and examines details such as usernames, timestamps, source addresses, event types, and destinations. Rules and analytics are used to identify suspicious behavior.

Event Correlation

SIEM connects related events that may appear harmless when viewed separately. For example, repeated failed logins followed by a successful login from an unfamiliar address may indicate a brute-force attack or compromised account.

Alert Generation

When events match detection rules or deviate from expected behavior, the SIEM generates alerts. Alerts may identify suspicious login attempts, unusual outbound connections, malware activity, or unauthorized access to sensitive resources.

Risk Prioritization

Analysts prioritize alerts according to severity, confidence, asset importance, and potential business impact. Activity involving a critical database or an administrator account may require faster attention than a low-risk event.

Investigation and Response

Analysts use correlated logs and timelines to identify the source, scope, and likely cause of an incident. Depending on available integrations and approval procedures, the SIEM may support actions such as disabling an account or isolating a device.

Reporting and Examples

SIEM dashboards and reports summarize incidents, alert trends, and response activities. Common examples include Splunk Enterprise Security, IBM QRadar, and Elastic Security.

Conclusion

SIEM improves centralized security monitoring by collecting events, correlating related alerts, prioritizing risks, and supporting faster incident response.

3. Security Monitoring and Anomaly Detection [10 Marks]

Introduction

A sudden increase in outbound traffic from a database server that normally transfers little data is an anomaly. It may result from a legitimate backup or update, but it could also indicate unauthorized access or data exfiltration.

Identify the Anomaly

Compare current traffic volume, timing, destination, and protocol with historical baselines. Confirm whether the increase occurred outside normal working hours and whether any planned maintenance or backup explains it.

Analyze Network Traffic

Use tools such as Wireshark or Zeek to examine connections, destination IP addresses, ports, protocols, and transfer patterns. Look for repeated connections to unfamiliar hosts or unusually large outbound transfers.

Review System and Database Logs

Check database queries, account activity, authentication logs, file exports, scheduled jobs, and administrator actions. Identify whether a user or process accessed sensitive information shortly before the traffic increase.

Check External Destinations

Compare destination domains and IP addresses with approved business services and threat-intelligence information. Unrecognized destinations or known malicious indicators require further investigation.

Assess Possible Data Exfiltration

Determine what information may have been accessed, whether it left the network, the volume transferred, and which account or process initiated the transfer. Preserve relevant logs and evidence for further analysis.

Containment

If malicious activity is suspected, follow the incident-response process to block the destination, restrict outbound traffic, disable compromised credentials, or isolate the server when appropriate. Ensure that containment does not unnecessarily disrupt critical services.

Prevention and Monitoring

Apply least-privilege access, database activity monitoring, firewalls, data loss prevention (DLP), encryption, and outbound traffic filtering. Configure alerts for unusual transfer volumes, new destinations, and transfers occurring outside expected hours.

Conclusion

Baseline monitoring, log investigation, network analysis, and suitable preventive controls help detect potential data exfiltration and reduce the chance of sensitive information leaving the organization.

4. Tools Used in the IT Industry for Threat Hunting [10 Marks]

Introduction

When malware is suspected on employee workstations, security teams should combine SIEM, endpoint detection and response (EDR), network analysis, vulnerability scanning, and threat-intelligence tools. Each provides a different view of the incident.

SIEM Tools – Splunk or Elastic Security

These tools collect and correlate logs from endpoints, servers, firewalls, and authentication systems. They help identify suspicious logins, repeated alerts, unusual processes, and communication patterns across multiple devices.

Endpoint Detection and Response – Microsoft Defender for Endpoint

EDR tools monitor processes, files, registry changes, and endpoint behavior. They can help investigate malware execution, identify affected workstations, trace activity, and isolate compromised devices where supported and authorized.

Network Analysis – Wireshark

Wireshark captures and inspects network packets. Analysts can examine suspicious connections, protocols, DNS requests, and data-transfer patterns to understand how infected devices communicate with external servers.

Vulnerability Scanning – Nessus

Nessus scans systems for missing patches, insecure configurations, and known vulnerabilities. These findings help determine how malware may have entered the environment and which weaknesses should be fixed.

Threat Intelligence – VirusTotal

VirusTotal can help check file hashes, domains, and IP addresses against multiple security sources. Results should be treated as investigative evidence and verified using internal context rather than as the sole basis for a decision.

Investigation and Containment

Collect relevant logs, identify affected endpoints, correlate indicators, and determine the likely infection path. Isolate confirmed infected devices, block malicious indicators, and preserve evidence before removing malware.

Recovery and Prevention

Remove malicious files and persistence mechanisms, patch vulnerabilities, reset exposed credentials, and restore systems from trusted sources if needed. Continue monitoring for repeated indicators or reinfection.

Conclusion

Using these tools together provides visibility into endpoint activity, network communications, vulnerabilities, and threat indicators, enabling a more complete investigation and response.

5. Integrated Threat Hunting and Incident Reporting [10 Marks]

Introduction

Integrated threat hunting combines log analysis, endpoint investigation, network evidence, and threat intelligence to investigate related indicators such as suspicious logins, malicious-domain connections, and unknown processes.

Data Collection

Collect authentication logs, endpoint alerts, DNS records, firewall logs, proxy logs, server events, and relevant network traffic. Record timestamps and preserve evidence so events can be compared accurately.

Log Correlation

Use Splunk or the ELK Stack to search and correlate events by user, host, IP address, domain, process, and time. For example, a suspicious login followed by an unknown process and a connection to a malicious domain may indicate a coordinated attack.

Threat Intelligence Verification

Check suspicious domains, IP addresses, URLs, and file hashes using a trusted threat-intelligence platform. Compare the results with internal evidence and consider that indicators may be outdated or shared by legitimate services.

Threat Investigation

Build a timeline of events, identify the account and endpoint involved, examine process trees and persistence mechanisms, and determine whether other systems show similar indicators. Establish the likely entry point and scope of the incident.

Malware Analysis

If an unknown file is suspected to be malicious, calculate its hash and examine its metadata and behavior in a controlled, isolated environment. Avoid running suspicious files on production systems, and escalate complex analysis to trained specialists.

Risk Assessment

Evaluate the sensitivity of affected systems, privileges of compromised accounts, potential data exposure, number of impacted devices, and business disruption. Assign a severity level to guide the response priority.

Containment and Eradication

Isolate compromised endpoints, block confirmed malicious domains, disable or reset affected accounts, and restrict suspicious network traffic. Remove malware, close exploited vulnerabilities, and preserve evidence according to organizational procedures.

Recovery and Validation

Restore systems from trusted backups where necessary, apply patches, rotate exposed credentials, and verify that systems operate normally. Continue monitoring for the same indicators to confirm that the attacker has not returned.

Threat-Hunting Report

Prepare a report containing the incident summary, objective, data sources, tools used, timeline, indicators of compromise, affected assets, evidence, risk rating, actions taken, and unresolved questions. Separate confirmed findings from assumptions.

Remediation Recommendations

Recommend multi-factor authentication, least-privilege access, regular patching, endpoint protection, DNS and outbound traffic filtering, SIEM detection-rule improvements, user awareness training, and periodic threat hunts.

Conclusion

A structured threat-hunting process supports accurate detection, evidence-based risk assessment, coordinated containment, and clear reporting. The lessons learned should be used to strengthen future monitoring and incident response.